§9 Limitations, open problems, outlook for Phases 2 and 3 — part 2 of 3 (delegations to Phase 1.α; open problems)

Source: `paper/section9_limitations.tex`, lines 467–846 (`sec:limitations_and_outlook`: `subsec:phase1a`, `subsec:open_problems`), at commit `e6e12cd`. Status: draft. Part 1 (1–466: introduction and L1–L18) is `09_limitations_1.md`; part 3 (847–1071: outlook for Phases 2 and 3, conclusion) is `09_limitations_3.md`.

Delegations to Phase 1.α

`Reference: subsec:phase1a`

What Phase 1.α is. In the paper, Phase 1.α is an auxiliary refinement. It keeps the premises of Phase 1 (deterministic, discrete time, compact spaces, non-branching, single-ACS focus) and the type of each object unchanged. For objects to which Phase 1 gave only a value range, a type or a carrier, it makes rules, thresholds and operational criteria concrete. A relaxation of the Phase 1 assumptions belongs to Phase 2, and the connection with ECIT belongs to Phase 3 (`subsec:phase_outlook`). Where the manuscript writes "Phase 1.α or Phase 2", the part that can be treated deterministically is considered in Phase 1.α and the rest in Phase 2.

The list below shows where the subjects for study lie. This paper does not state that any of the items will be formalized in Phase 1.α.

1. **Candidates for constitutively necessary functions (classification A):** the required-environment utilization function (`rem:env_utilization_function`) and the cross-cutting capability to respond to environmental change (`rem:env_change_response`).
2. **Composition of mechanisms:** a reformulation of TIPA that allows the mechanisms to be integrated (`obs:tipa_integration`); local responses of the PSL layer that arise without deliberate selection, in general, including involuntary local responses with no determined protective function, that is, those outside the scope of PSL-4 (`subsubsec:psl4`).
3. **Cognitive processing capability:** formalization of quantitative capacity, time responsiveness and quality of integration (`rem:cognitive_capacity_policy`).
4. **The three principles:** versions of the response-inability environment set, the response diversity and the conditional analysis that take into account the response of the action-response component (`rem:same_time_alternatives`); the long-term aspects of FR (`rem:sd_local_scope`); and a construction of the non-implications among the time-indexed satisfaction predicates by models that satisfy the assumptions and constitutive requirements of §2 (`rem:principle_implication_scope`).
5. **Content of TIPA:** enumeration and classification of the qualities of a request (`def:request_signal_types`); the relation between means and purposes (`rem:purpose_hierarchy`); the distinction among reasons for ending a purpose without achieving it (`def:purpose_status`); tracing the origin of a bypass or replacement of the evaluation mechanism (`rem:self_attribution_scope`); and the correspondence between internal object tokens and external slots or individuals (`rem:Eext_decomposition_reading`).
6. **Time-indexed quantities and records:** the computation rule for the effective development freedom (`def:deff_section5`); a mechanism-by-mechanism description of how a bias is formed and the identification of its attribution division (`rem:bias_formation_vs_attribution`); and conditions on the detailed type of Update under which retained records agree (`def:s5_record_agreement`, `rem:update_type_two_layers`).
7. **Relations:** the items of `rem:phase1a_delegation` in §6; concrete forms and finite approximations of the local size indicator (`rem:finite_window_diff_limit`); and a quantitative rule for judging the principal direction (`rem:dir_undefined`).
8. **Organizational ACS:** descriptions that take an organizational ACS as a relation object, and the application of bidirectional composition (`rem:org_terms_nonimplication`).

Open problems

Reference: subsec:open_problems

These are research problems that Phase 1 does not resolve. Each indicates a future direction. The paper does not claim that any of them is solvable, that its solution is unique, or that progress will follow as a matter of course. The section contains eleven problems. They are summarized below in the manuscript order; none of the statements is a result.

Reformulating individual identity when branching is permitted (prob:branching_identity). *Background:* ass:no_branching (a strong assumption) allows at most one realization at time $t+1$ to be associated with an individual realization at time t as its continuation, which excludes copies and multiple successor realizations. The structural self $S_{\mathrm{struct}}(\Sigma_i)$ (def:s_struct) and the preliminary conditions of temporal self-identity (def:time_identity) are described on this premise. The splitting of an organizational ACS is included, as the case of several successors of one organization. *Questions:* (i) a formalization of an identity relation that decides which successor is taken as the continuation of the same individual, whether several realizations are treated as continuations on one lineage, or whether each realization after branching is a separate individual; (ii) a reformulation of S_{struct} and of the preliminary conditions of temporal self-identity when branching is allowed, and the treatment of collapse Col_t (§7) under branching; (iii) the logical and dynamical relation of branching to blockage $A_t = \emptyset$ and collapse $\mathrm{Col}_t = 1$, and how to describe the persistence of a population or lineage that includes several realizations. *Barrier:* a new formal identity relation for persistence is needed. It is a condition on individual identity separate from the single-valuedness of f; Phase 1 places non-branching as an assumption apart from f. *Direction (Phase 2 or later):* candidates are temporal continuation relations between realizations, set-valued correspondences or lineage structures, aiming at a framework in which the non-branching identity under ass:no_branching is recovered as a special case.

Calibration and predictive interpretation of T-STAB parameters (prob:stability). For $\underline{s}_{\mathrm{SM}}$, $\delta_{\max}$ and W_T, which T-STAB takes as given, the problem is to clarify quantitative calibration conditions consistent with the kind of implementation of the ACS. Four subjects are to be studied: (1) sensitivity to the choice of the reference scale and the lower reference value of the SM evaluation; (2) robustness to the window width W_T and the fluctuation bound $\delta_{\max}$; (3) identifiability of connections, under additional conditions, between T-STAB violation and blockage, collapse or warning conditions; (4) parameter estimation from implementation data and its validation. The T-STAB definition of Phase 1 gives neither a general calibration rule nor predictive performance.

Quantitative identification of estimation errors and internal update rules (prob:estimation_error). For the M1 estimation error, the M2 evaluation error, the internal-model update, the bias strengths $\boldsymbol{\beta}_t$ of the individual biases (including the specification of the reference criterion and scale of each bias), $d_{\mathrm{eff},t}$, $\pi_{\tau,t}$ and the four health quantities, the problem is to identify the readout rule, the transition of the carrier and the propagation of error in each implementation, and to evaluate their effect on T-STAB, the feasible action set and reaching of blockage. A general convergence rate or error distribution cannot be derived from the abstract type-preserving conditions of Phase 1, so additional models and data are needed.

Conditions for acquiring kernel-maintaining responses (prob:kernel_response_acquisition_conditions). *Background:* prop:kernel_maintenance shows the following. If $x_T \in \mathrm{Viab}_f(K)$ and, for every $s \geq T$, $e_s \in E_{\mathrm{req}}(x_s)$ and $u_s \in U^{\mathrm{Viab}}(x_s)$ hold, then for every $s \geq T$ both $x_s \in \mathrm{Viab}_f(K)$ and $A_s \neq \emptyset$ hold. The condition $u_s \in U^{\mathrm{Viab}}(x_s)$ for all $s \geq T$ may or may not hold, depending on the mechanism that generates the effective action (§4) and on the update of the internal model (§5) (def:kernel_response_acquisition). This paper does not state as a general consequence that acquisition occurs, that the update of the internal model leads to acquisition, or that the action actually chosen belongs to $U^{\mathrm{Viab}}(x)$ from $U^{\mathrm{Viab}}(x) \neq \emptyset$ (rem:kernel_response_scope). It does not require an ACS to hold or compute $\mathrm{Viab}_f(K)$ or $U^{\mathrm{Viab}}(x)$ internally (subsec:description_levels_global). *Questions:* (i) identify conditions under which observation (P_{sens}) and internal estimation retain the information needed to distinguish kernel-maintaining responses from the others; (ii) examine under which additional conditions on the update of the internal model and on the mechanisms forming and selecting responses there exists a time T such that $E_{\mathrm{req}}(x_s) \neq \emptyset$ and $u_s \in U^{\mathrm{Viab}}(x_s)$ hold for every $s \geq T$ (acquisition as defined in def:kernel_response_acquisition), and whether an upper bound on that time can be given; (iii) organize the relation between the choice of kernel-maintaining responses and the realized trajectory staying in $\mathrm{Viab}_f(K)$, under

sequences of external environment states in which $e_s \in E_{\mathrm{req}}(x_s)$ is interrupted. *Barrier:* Phase 1 defines the viability kernel and the kernel-maintaining response as formal quantities but gives no general condition on how an internal mechanism approximates and selects them from internal information alone. *Direction:* conditions for deterministic finite realizations are a Phase 1.α refinement; extensions to general dynamics that include uncertainty of estimation and stochastic variation of the external environment are Phase 2.

From relation description to causal models (`prob:relational_causality`). While keeping the descriptive relation structure of §6, the problem is to clarify, only for the applications that need it, conditions under which a per-object causal contribution can be defined. The subjects are the effect on causal interpretation of intervention rules, comparison trajectories, counterfactuals, confounding among several relations, the choice of the time window and the choice of the size functional. The premise is that a descriptive relation category is not read as a causal category.

Observability of blockage, collapse and recovery (`prob:collapse_observability`). The problem is to clarify which observables, history lengths, access to internal states and models of external action are needed to judge blockage, collapse and recoverability from finite observation. It also asks for formulations of the prediction of the blockage reaching time and the collapse reaching time, of conditions of inflow into the path sets, of the robustness of the recovery judgment and of upper bounds on misjudgment.

Multi-step recoverability to K and required environmental factors (`prob:multistep_recovery`). *Background:* the one-step critical required factor, EU and local FR of §3 are one-step conditions (`rem:eu_one_step_scope`). Blockage at a time with $x_t \notin K$ only expresses that K cannot be reached in one step. This paper does not state as a general consequence that K cannot be reached in finitely many steps (`rem:blockage_scope_sec2` of §2). Since $\mathrm{Viab}_f(K) \subseteq K$, $x \notin \mathrm{Viab}_f(K)$ is no ground for the impossibility of multi-step recovery. *Questions:* (i) define reachability of K in finitely many steps from a state outside K, with the order of quantification over the action sequence and the exogenous background sequence stated (whether the action sequence is fixed before the background components or chosen according to the state and external environment state at each step); (ii) characterize the required environmental factors for that reachability, including their relation to the one-step critical required factors; (iii) describe separately blockage at a time in the middle of multi-step recovery and the impossibility of recovery in finitely many steps. *Barrier:* Phase 1 defines no multi-step reachability and limits its connection with required environmental factors to one-step conditions. *Direction:* the part that can be handled under deterministic state transitions and environment realization (the exogenous background sequence and g_O) is a Phase 1.α refinement; reachability with stochastic variation of the external environment is Phase 2.

Dynamics, identity and aggregation rules of organizational ACSs (`prob:social_dynamics`). On the basis of the time-indexed participating-ACS set of §8 and the descriptive applicability convention of the organizational ACS, Phase 2 is to build extensions that treat joining, leaving, generation and disappearance of participating ACSs; time change of the organizational boundary, the division of functions and the relation structure; conditions for retaining and regaining organizational identity; the treatment of organization-level EU and FR and a concrete form of T-STAB; and organization-level blockage and collapse paths. The aggregation from individual quantities to organizational quantities is model-dependent, and no single mean, minimum or weighted mean is assumed as a general rule.

Capability improvement and shrinkage of effective options (the capability paradox) (`prob:capability_paradox`). *Background:* the problem considers the hypothesis that an improvement of cognitive processing capability (`rem:cognitive_capacity_policy` of §2) widens the representability of the internal model and may connect with a wider range in which action candidates (such as Cand_t of §4) can be generated. On the other hand, the action input given to the state-transition map at each time is one $u_t \in U$ (`def:state_transition`, `rem:f_reading`), whatever the level of capability. The *capability paradox* is the view that, as capability improves, the management load for maintaining oneself and the constraints arising from relations with other ACSs and other external objects increase, so that effective options may be narrowed to a few when evaluated over a long time horizon. A further view is recorded as a candidate consequence of the capability paradox: a longer prospective time horizon may favor the retention of coordinative actions maintaining the feasibility of ACSs with one another among effective options. These are hypotheses of the paper and are not asserted as general consequences of the definitions and propositions of Phase 1; the management load and the effectiveness are not formally defined in Phase 1. *Questions:* (i) distinguish the latent options (action candi-

dates the internal model can generate) from the effective options (the set of actions regarded as effective once a time horizon is specified). A candidate criterion of effectiveness is the multi-step extension of the one-step feasible action set A_t (`def:at`). However, once the initial value, the exogenous background sequence and g_O are fixed under deterministic transitions, the first T steps of an action sequence that keeps feasibility for $T+1$ steps form an action sequence that keeps feasibility for T steps; hence the set that the first action of such sequences can take does not grow when the horizon is lengthened. This shrinkage arises from the definition, whatever the level of capability. The content of the capability paradox therefore lies not in whether this shrinkage occurs but in how an improvement of capability changes its degree. (ii) Formalize the effect that an improvement of capability has on the set of effective options through changes in quantitative capacity, time responsiveness and quality of integration (`rem:cognitive_capacity_policy`), relations with external objects (§6) and so on, including effects that work in the opposite direction, such as an improvement of capability widening the feasible action set or changing the constraints themselves. (iii) Formulate the relation between the length of the time-horizon outlook and the persistence of feasibility-maintaining actions among ACSs as effective options, in situations where several ACSs are part of one another's external environment. That coordinative actions may be among effective options does not by itself give a condition excluding non-coordinative actions from them; the problem therefore asks for the conditions under which effective options are narrowed to coordinative actions. *Barrier:* Phase 1 centers on the deterministic description of a single ACS and formalizes neither the simultaneous time evolution of several ACSs, nor a criterion of effectiveness over a time horizon, nor the mutual dependence of the feasibility of ACSs. The relation description of §6 contains no causal contribution, and §8 is limited to a restricted structural application at the organizational level. *Direction (Phase 2 or later):* Phase 2 formalizes the interaction of several ACSs and the set of effective options over a time horizon together with the extension in `prob:social_dynamics`. Phase 3 examines, within `prob:ecit_full`, the conditions under which effective options are narrowed to coordinative actions and their relation to the coordination concepts of ECIT. At no stage is the capability paradox adopted as a premise; both the conditions under which it holds and those under which it does not are objects of study.

Cross-domain correspondence and empirical validation (`prob:cross_domain`). For humans and organisms, AI and robots, organizations and so on, the problem is to fix operational correspondences for the 7-tuple of the ACS, the three principles, TIPA, PSL and ISL, the variables of §5, the relation descriptors, and blockage and collapse, and to evaluate measurability, reproducibility, predictive performance and falsifiability. Formal examples of correspondence must be distinguished from empirically supported correspondences.

Formal connection with ECIT (`prob:ecit_full`). The problem is to construct a formal correspondence between the structure of feasibility preservation given by ACS theory and the concepts of ECIT such as ethical coordination, love, wisdom and responsibility. It is not assumed that a normative value judgment follows from a descriptive feasibility condition. For the results on organizational dynamics in Phase 2 treated in `prob:social_dynamics`, the connection with ECIT is to be examined while keeping their position as descriptive results.

Dependencies among the problems

`Reference: rem:problem_dependencies`

The problems have practical dependencies, but the paper does not claim a strict logical order. For a single ACS, the basic subjects are temporal self-identity, the calibration of T-STAB, estimation error, internal update rules, the conditions for acquiring kernel-maintaining responses, and multi-step recoverability. One natural order of study on top of these is: single ACS → relations and blockage → organizational and social dynamics → connection with ECIT. `prob:capability_paradox` spans both organizational and social dynamics and the connection with ECIT. Cross-domain empirical validation can proceed in parallel with each stage.

Check status. This summary adds no new definition, assumption or result. The delegations to Phase 1.α are subjects for study and the open problems are unresolved; neither is stated here as a result. Labels in backticks are shared with the manuscript. It has been checked against the manuscript by the drafting model and audited by an external AI model. It remains a draft until the author checks it against the manuscript. No independent human third-party verification is claimed.